Creating and using RDS.

1. What is RDS?

Amazon RDS

Amazon RDS (Relational Database Service) is a **managed service by AWS** that lets you run relational databases without handling infrastructure.

In simple terms:
You focus on data → AWS handles server, OS, maintenance.

2. Why RDS Exists (Core Idea)

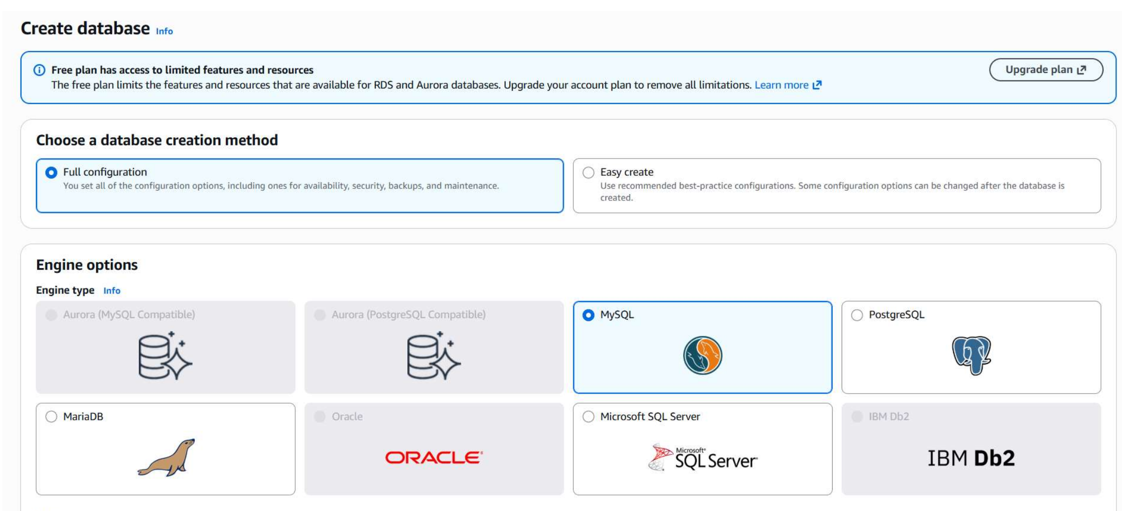
Without RDS:

- Install DB manually
- Configure backups
- Handle crashes
- Patch OS

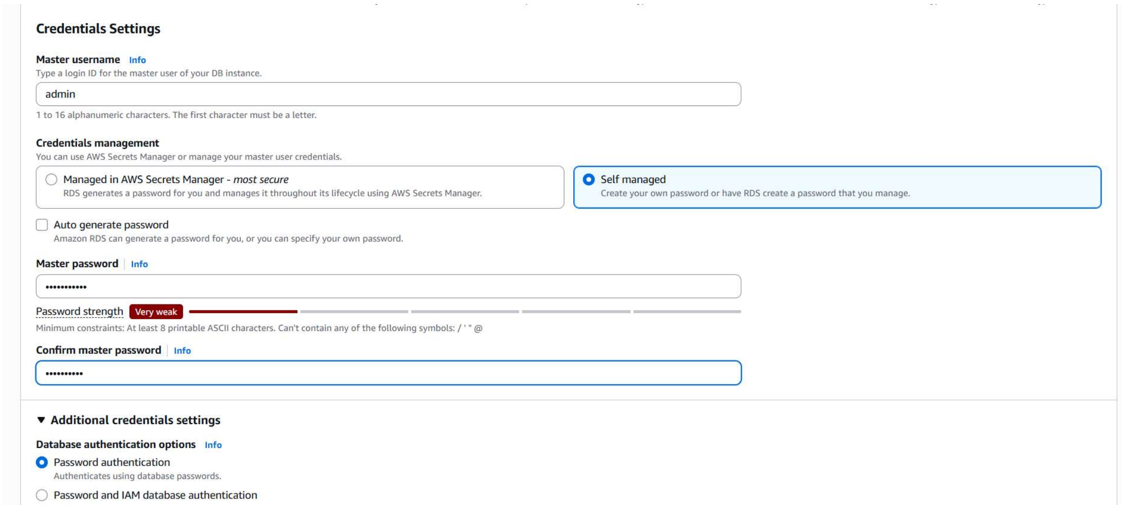
With RDS:

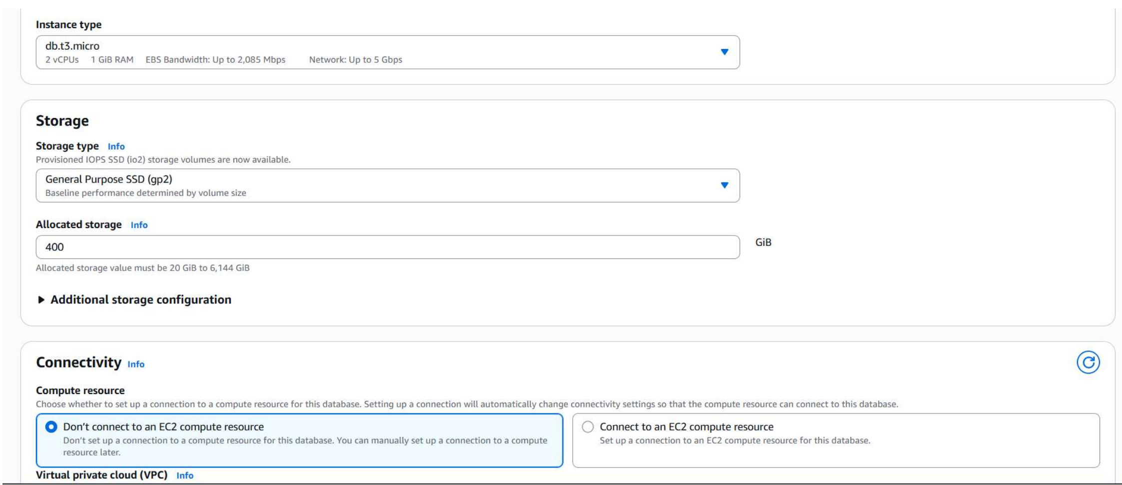
- AWS automates all of it

Creation Of RDS.

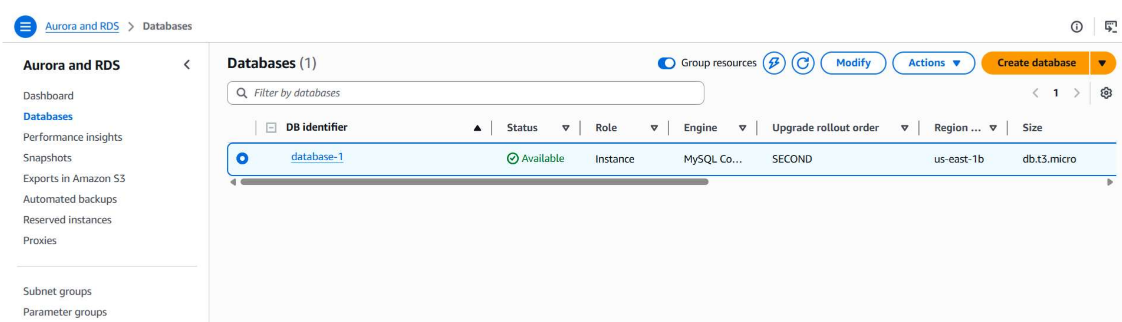


Then creation of db name , username and password.





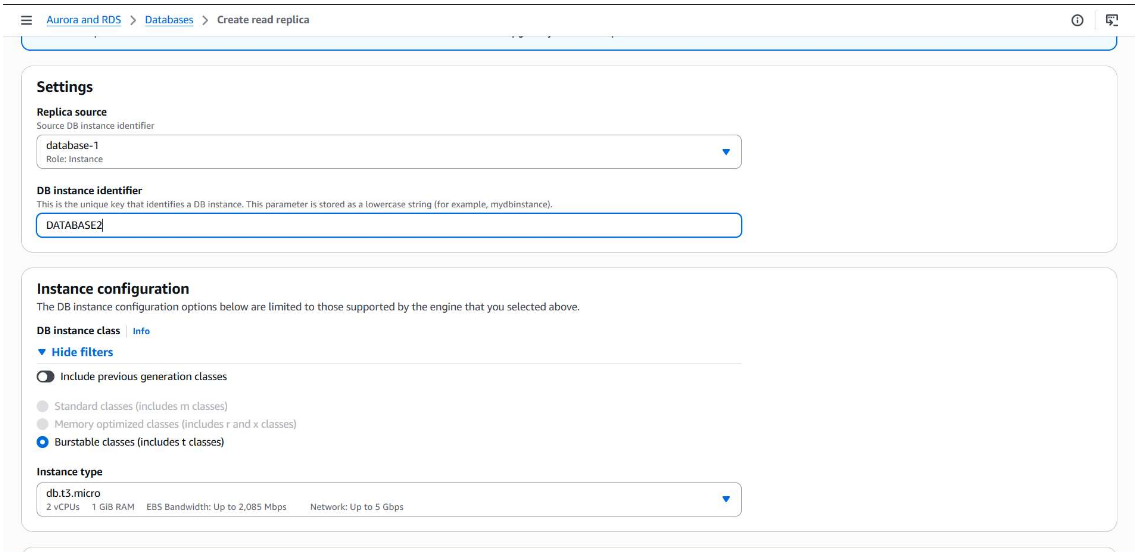
Created Database.



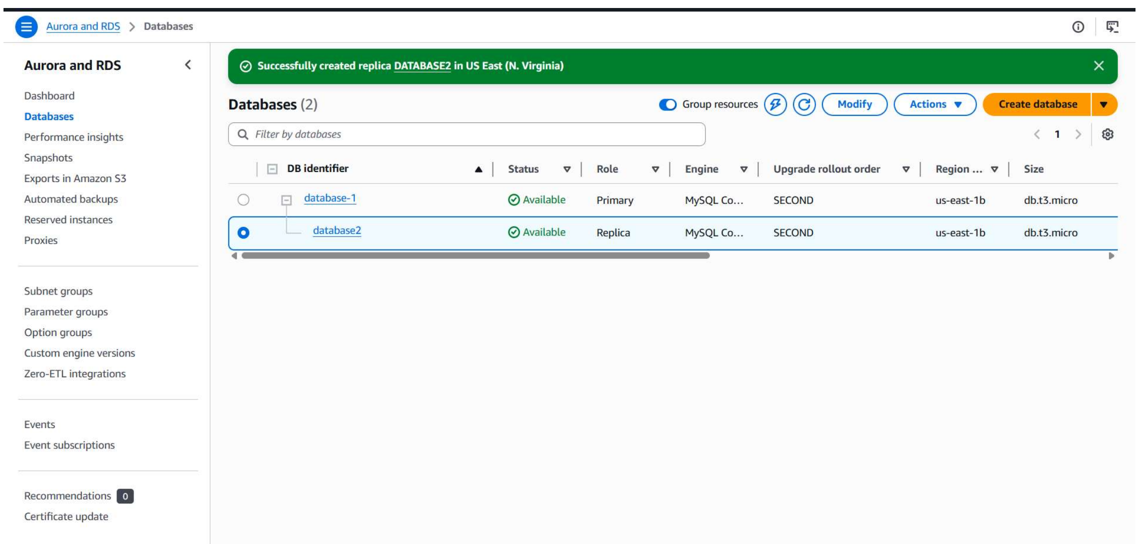
Read Replica.

Then we have to Create an Read Replica.

It Will allow you to duplicate your data when it acts as secondary database when primary db fails it acts as an primary database.



This is the Created Read Replica.



Then we have to connect to Database through the instance.

Command -- **mysql -h your-endpoint.rds.amazonaws.com -u admin -p.**

```
Setting up libhttp-message-perl (6.45-1ubuntu1) ...
Setting up mysql-server (8.0.45-0ubuntu0.24.04.1) ...
Setting up libcgi-pm-perl (4.63-1) ...
Setting up libhtml-template-perl (2.97-2) ...
Setting up libcgi-fast-perl (1:2.17-1) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.7) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@ip-172-31-21-232:/home/ubuntu# mysql -h database-1.cq56uimsqe8t.us-east-1.rds.amazonaws.com -P 3306 -u admin -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 30
Server version: 8.4.7 Source distribution

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql>
```

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Then I have created one database and Created one table.

```
mysql> SHOW DATABASES;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
4 rows in set (0.01 sec)

mysql> CREATE DATABASE HARIDB;
Query OK, 1 row affected (0.04 sec)

mysql> USE HARIDB;
Database changed
```

```
mysql> USE HARIDB;
Database changed
mysql> CREATE TABLE Persons (
    ->     PersonID int,
    ->     LastName varchar(255),
    ->     FirstName varchar(255),
    ->     Address varchar(255),
    ->     City varchar(255)
    -> );
Query OK, 0 rows affected (0.05 sec)
```

```
mysql> DESC Persons;
+-----------+--------------+------+-----+---------+-------+
| Field     | Type         | Null | Key | Default | Extra |
+-----------+--------------+------+-----+---------+-------+
| PersonID  | int          | YES  |     | NULL    |       |
| LastName  | varchar(255) | YES  |     | NULL    |       |
| FirstName | varchar(255) | YES  |     | NULL    |       |
| Address   | varchar(255) | YES  |     | NULL    |       |
| City      | varchar(255) | YES  |     | NULL    |       |
```

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